Parametric Kleisli Categories

Cartesian Cat $\mathcal{C}$

Objects a, b, c $\qquad \Delta_a : a \to a \times a$

morphisms f, g, h $\qquad \delta_a : a \to 1$

composition $f \circ g$

Tensor $f \times g$

Probability Monad

$P : \mathcal{C} \to \mathcal{C}$

$\mu : P^2 \to P$

$\eta : 1 \to P$

$\nabla_{x,y} : Px \times Py \to P(x \times y)$

Markov Cat $Kl(P)$

morphism $\underset{\sim}{f} : a \to b$ in $Kl(P)$

corresponds to $f : a \to Pb$ in $\mathcal{C}$ $\qquad \underset{\sim}{id}_a = \eta_a$

$a \otimes b$ in $Kl(P)$ is $a \times b$ in $\mathcal{C}$

$\underset{\sim}{f} \otimes g$ in $Kl(P)$ $\qquad f : a \to Pb \quad g : x \to Py$

$$a \times x \xrightarrow{f \times g} Pb \times Py \xrightarrow{\nabla_{b,y}} P(b \times y)$$

$\underset{\sim}{\Delta}_x : x \to x \otimes x$ $\qquad x \xrightarrow{\Delta_x} x \times x \xrightarrow{\eta_{x \times x}} P(x \times x)$

$\underset{\sim}{\delta}_x : x \to 1$ $\qquad x \xrightarrow{\delta} 1 \xrightarrow{\eta_1} P1$

$CoKl(C \otimes -)$

$\overset{CoKl(C\otimes -)}{f^\# : x \to y}$ is $\overset{Kl(P)}{\underset{\sim}{f} : C \otimes x \to y}$ is $\overset{\mathcal{C}}{f : C \times x \to Py}$

$id_x^\# : x \to x$ is $\underset{\sim}{\varepsilon}_x : C \otimes x \to x$

$\underset{\sim}{\delta}_c \otimes \underset{\sim}{id}_x =$

Biklesli

id =

$$\left[(\delta_c \circ \eta_1) \times (\eta_x) \right] \circ \nabla_{1x}$$

$$\left.\begin{array}{l} c \xrightarrow{\delta_c} 1 \xrightarrow{\eta_1} P1 \\ x \xrightarrow{\eta_x} Px \end{array}\right\} \xrightarrow{\nabla_{1x}} P(1 \times x) \simeq Px \overset{?}{=} C \times X \xrightarrow{\delta_c \times id_x} X \xrightarrow{\eta_x} PX$$

$$\left.\begin{tikzcd} C \arrow[r, "\delta_c"] & 1 \arrow[r, "\eta_1"] & P1 \\ X \arrow[rr, "\eta_X"'] & & PX \end{tikzcd}\right\} \xrightarrow{\nabla_{1X}} P(1\times X) \simeq PX \overset{?}{=} C\times X \xrightarrow{\delta_c \times id_X} X \xrightarrow{\eta_X} PX$$

$$\left.\begin{tikzcd} C \arrow[r, "\delta_c"] & 1 \arrow[r, "\eta_1"] & P1 \\ X \arrow[r, "id_X"'] & X \arrow[r, "\eta_X"'] & PX \end{tikzcd}\right\} \xrightarrow{\nabla_{1X}} P(1\times X) \simeq PX$$

$$\underbrace{\qquad\qquad\qquad}_{\eta_{1\times X}}$$

Composition

$\mathrm{CoKl}(c\otimes -)$	$\mathrm{Kl}(P)$
$X \xrightarrow{f^\#} Y \xrightarrow{g^\#} Z$	$C\otimes X \xrightarrow{\underset{\sim}{\Delta}_c \otimes \underset{\sim}{id}_X} C\otimes C\otimes X \xrightarrow{\underset{\sim}{id}_C \otimes \underset{\sim}{f}} C\otimes Y \xrightarrow{\underset{\sim}{g}} Z$

$f^\# : X \to Y$

$\underset{\sim}{f} : C\otimes X \to Y$

$f : C\times X \to PY$

$$C \xrightarrow{\Delta_c} C\times C \xrightarrow{\eta_{C\times C}} P(C\times C)$$

$\underset{\sim}{\Delta}_C = \Delta_c \mathbin{⨾} \eta_{C\times C}$ $\qquad$ $\underset{\sim}{id}_C = \eta_C : C \to PC$ $\qquad$ $\underset{\sim}{g} = g : C\times Y \to PZ$

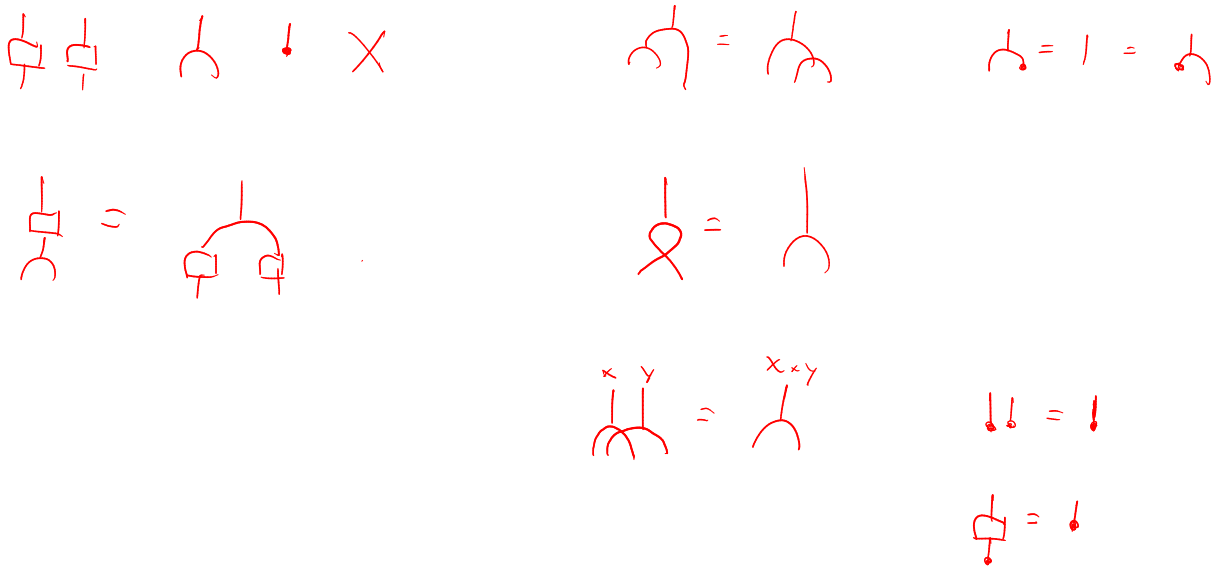
$\underset{\sim}{id}_X = \eta_X : X \to PX$ $\qquad$ $\underset{\sim}{f} = f : C\times X \to PY$

$$\underset{\sim}{\Delta}_C \otimes \underset{\sim}{id}_X = [(\Delta_c ; \eta_{C\times C}) \times \eta_X] ; \nabla_{C\times C, X}$$

$$\begin{tikzcd} C \arrow[r, "\Delta_c"] & C\times C \arrow[r, "\eta_{C\times C}"] & P(C\times C) \arrow[r, "\nabla_{C\times C, X}"] & P(C\times C\times X) \\ X \arrow[rr, "\eta_X"'] & & PX & \end{tikzcd}$$

Starting over

Cartesian Base

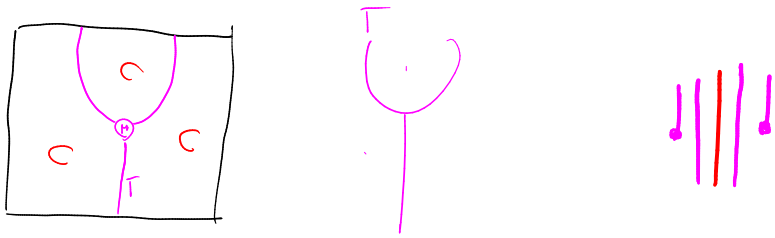


With Monad

$T : \mathcal{C} \to \mathcal{C}$

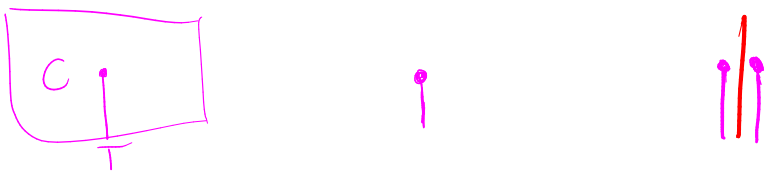
$\mu : T^2 \to T$

$$\begin{tikzcd} C \arrow[r, "T"] \arrow[rr, bend right, "T"'] & C \arrow[r, "T"] \arrow[d, phantom, "\Downarrow \mu"] & C \\ & & \end{tikzcd}$$



$\eta : 1 \to T$

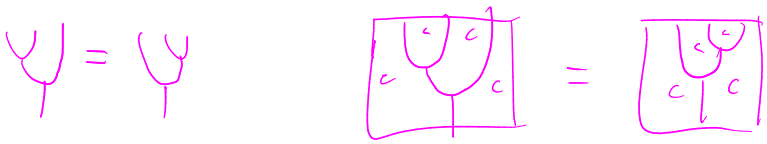
$$\begin{tikzcd} C \arrow[r, bend left, "1"] \arrow[r, bend right, "T"'] \arrow[r, phantom, "\Downarrow \eta"] & C \end{tikzcd}$$



Wire Naturality:

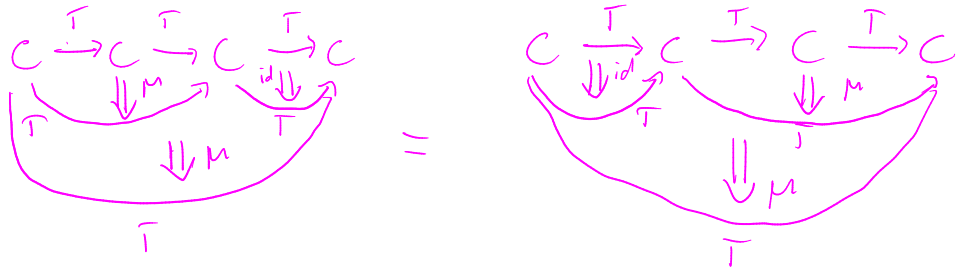
$$\begin{tikzcd} T^2a \arrow[r, "\mu_a"] \arrow[d, "T^2f"'] & Ta \arrow[d, "Tf"] \\ T^2b \arrow[r, "\mu_b"'] & Tb \end{tikzcd} \qquad \begin{tikzcd} a \arrow[r, "\eta_a"] \arrow[d, "f"'] & Ta \arrow[d, "Tf"] \\ b \arrow[r, "\eta_b"'] & Tb \end{tikzcd}$$

Monad Laws

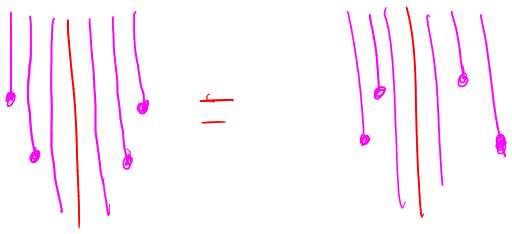


Associativity

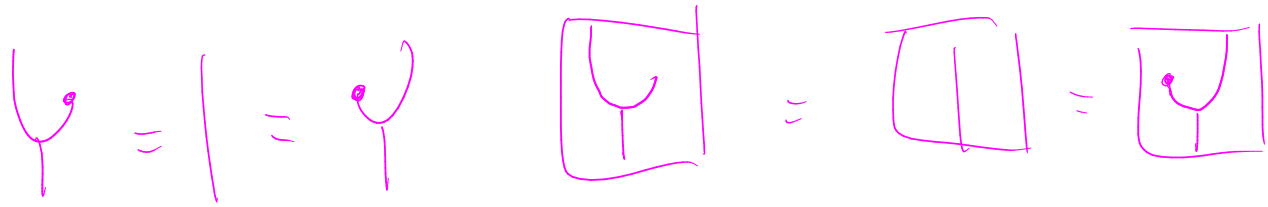
$$\begin{tikzcd} T^3 \arrow[r, "T\mu"] \arrow[d, "\mu T"'] & T^2 \arrow[d, "\mu"] \\ T^2 \arrow[r, "\mu"'] & T \end{tikzcd}$$



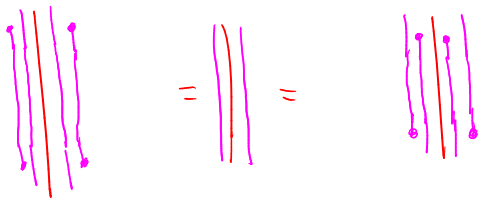
$$\begin{tikzcd} T^3X \arrow[r, "T\mu_X"] \arrow[d, "\mu_{TX}"'] & T^2X \arrow[d, "\mu_X"] \\ T^2X \arrow[r, "\mu_X"'] & TX \end{tikzcd}$$



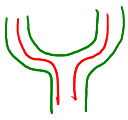
Unitality



$$\begin{tikzcd} T \arrow[r, "T\eta"] \arrow[d, "\eta T"'] \arrow[dr, equal] & T^2 \arrow[d, "\mu"] \\ T^2 \arrow[r, "\mu"'] & T \end{tikzcd} \qquad \begin{tikzcd} TX \arrow[r, "T\eta_X"] \arrow[d, "\eta_{TX}"'] \arrow[dr, equal] & T^2X \arrow[d, "\mu_X"] \\ T^2X \arrow[r, "\mu_X"'] & TX \end{tikzcd}$$



Laxator

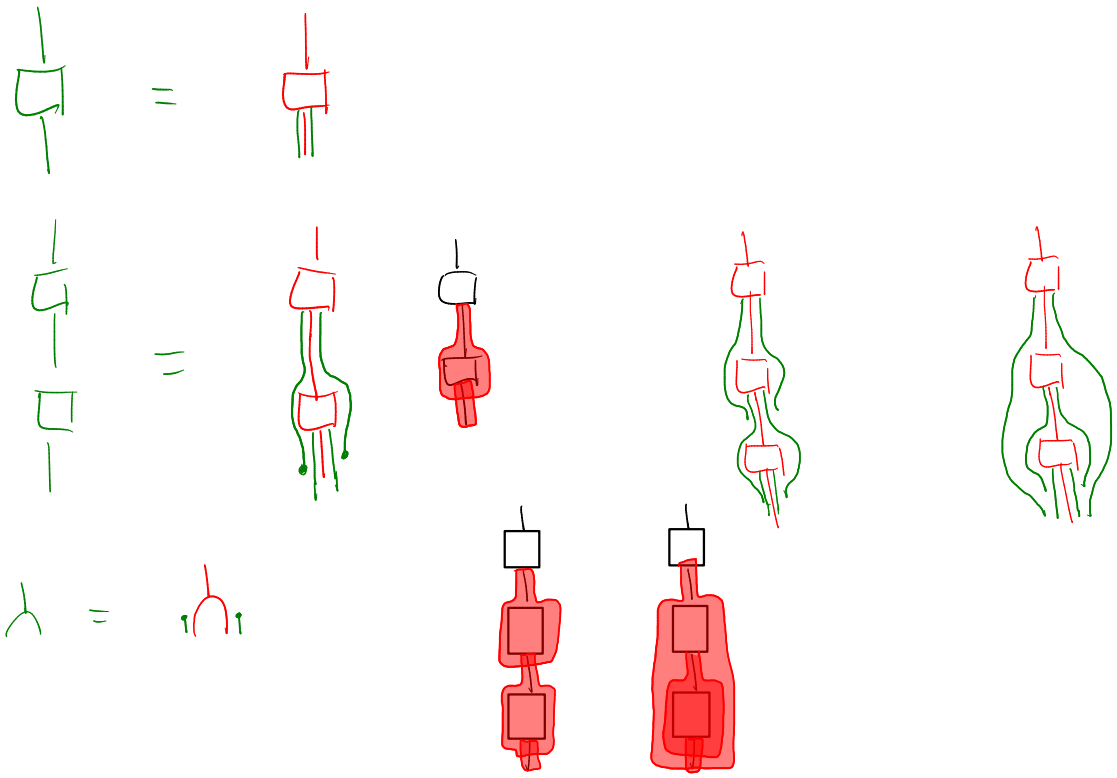


monoidal NTs

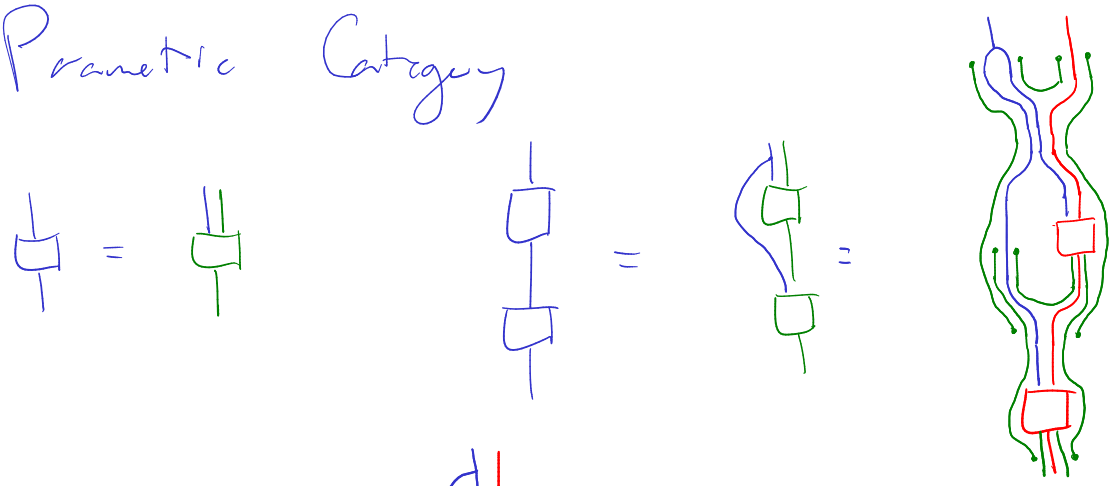
$TTX \otimes TTY \longrightarrow$

Unitor

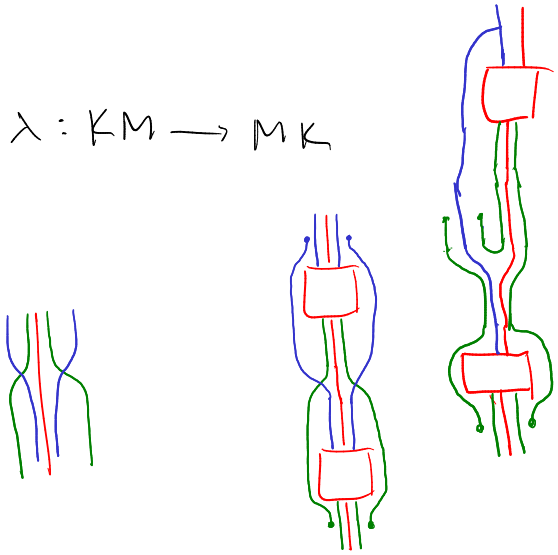
Kleisli Category



Pramatic Category



$\lambda : KM \longrightarrow MK$

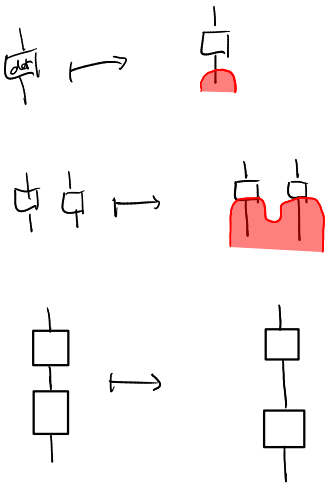


$\lambda : C \times MX \xrightarrow{1 \times MX} MC \times MX \xrightarrow{\nabla} M(C \times X)$

Cokl(i ⊙ −)

Kl(P)

Kl(P) to Base Rules



$$\begin{tikzcd} C \arrow[r, bend left=60, "F"{name=U}] \arrow[r, bend right=60, "F'"'{name=D}] \arrow[Rightarrow, from=U, to=D, "\alpha"] & D \arrow[r, bend left=60, "G"{name=U}] \arrow[r, bend right=60, "G'"'{name=D}] \arrow[Rightarrow, from=U, to=D, "\beta"] & E \end{tikzcd}$$

$x \in C$ $\qquad$ $\alpha_x : Fx \longrightarrow F'x$

$f : x \to y$

$$\begin{tikzcd} Fx \arrow[r, "Ff"] \arrow[d, "\alpha_x"'] & Fy \arrow[d, "\alpha_y"] \\ F'x \arrow[r, "F'f"'] & F'y \end{tikzcd}$$

$u \in D$ $\qquad$ $\beta_u : Gu \longrightarrow G'u$

$g : u \to v$

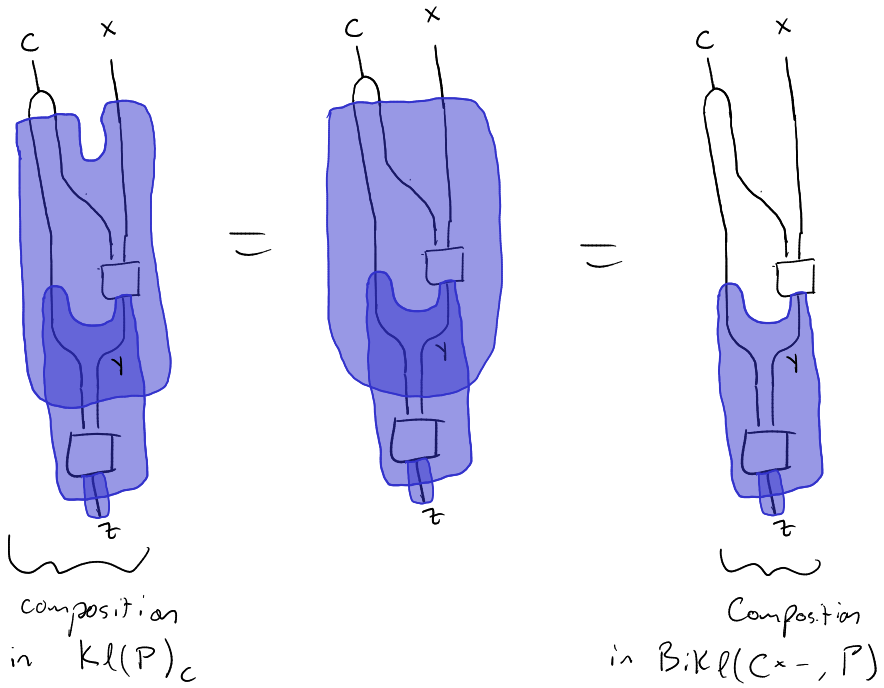
$\alpha_x : Fx \to F'x$

$$\begin{tikzcd} Gu \arrow[r, "Gg"] \arrow[d, "\beta_u"'] & Gv \arrow[d, "\beta_v"] \\ G'u \arrow[r, "G'g"'] & G'v \end{tikzcd}$$

$\alpha \mid \beta : F \mathbin{;} G \longrightarrow F' \mathbin{;} G'$

$(\alpha \mid \beta)_x : GFx \longrightarrow G'F'x$

$$\begin{tikzcd} GFx \arrow[r, "G\alpha_x"] \arrow[dr, "\beta_{Fx}"'] & GF'x \arrow[r, "\beta_{F'x}"] & G'F'x \\ & G'Fx \arrow[ur, "G'\alpha_x"'] & \end{tikzcd}$$



$Kl(P)_c$

$BiKl(c\times -, P)$

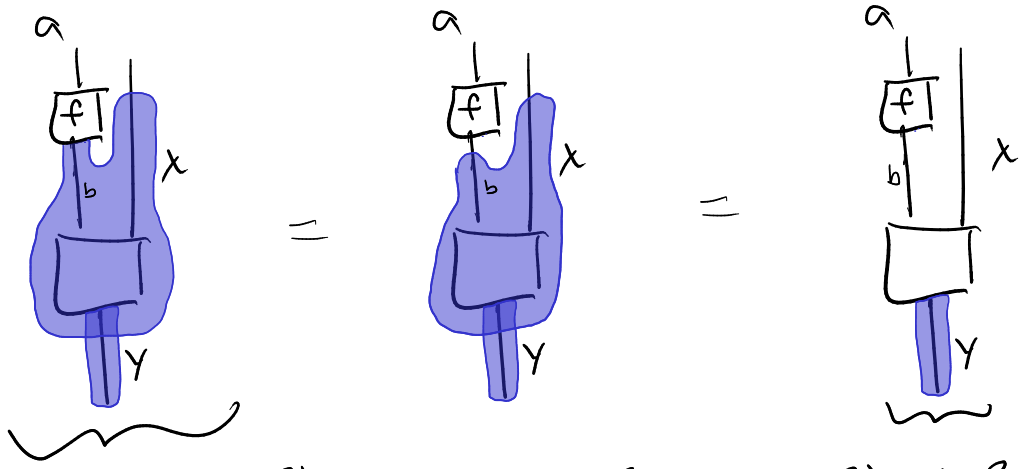
$x \to y$

$c \otimes x \to y$

$c \times x \to Py$

$c \times x \to Py$

$f : a \to b$ deterministic in $Kl(P)$



$f^* : Kl(P)_b \to Kl(P)_a$

$f^* : BiKl(b\times -, P) \to BiKl(a\times -, P)$